334. $\frac{j}{-6.2} = -3$

335. $-\frac{2}{3} = y + \frac{3}{8}$

336. $s - 1.75 = -3.2$

337. $\frac{11}{20} = -f$

338. $-3.6b = 2.52$

339. $-4.2a = 3.36$

340. $-9.1n = -63.7$

341. $r - 1.25 = -2.7$

342. $\frac{1}{4}n = \frac{7}{10}$

343. $\frac{h}{-3} = -8$

344. $y - 7.82 = -16$

Translate to an Equation and Solve

In the following exercises, translate and solve.

345. The difference of n and 1.9 is 3.4.

346. The difference n and 1.5 is 0.8.

347. The product of -6.2 and x is -4.96.

348. The product of -4.6 and x is -3.22.

349. The quotient of y and -1.7 is -5.

350. The quotient of z and -3.6 is 3.

351. The sum of n and -7.3 is 2.4.

352. The sum of n and -5.1 is 3.8.

Everyday Math

353. Shawn bought a pair of shoes on sale for $\$78$. Solve the equation $0.75p = 78$ to find the original price of the shoes, p.

354. Mary bought a new refrigerator. The total price including sales tax was $\$1{,}350$. Find the retail price, r, of the refrigerator before tax by solving the equation $1.08r = 1{,}350$.

Writing Exercises

355. Think about solving the equation $1.2y = 60$, but do not actually solve it. Do you think the solution should be greater than 60 or less than 60? Explain your reasoning. Then solve the equation to see if your thinking was correct.

356. Think about solving the equation $0.8x = 200$, but do not actually solve it. Do you think the solution should be greater than 200 or less than 200? Explain your reasoning. Then solve the equation to see if your thinking was correct.

Self Check

ⓐ After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.

I can...	Confidently	With some help	No-I don't get it!
determine whether a decimal is a solution of an equation.			
solve equations with decimals.			
translate to an equation and solve.			

ⓑ On a scale of 1–10, how would you rate your mastery of this section in light of your responses on the checklist? How can you improve this?

5.5 Averages and Probability

Learning Objectives

By the end of this section, you will be able to:

- Calculate the mean of a set of numbers
- Find the median of a set of numbers
- Find the mode of a set of numbers
- Apply the basic definition of probability

☑ **BE PREPARED** 5.13 Before you get started, take this readiness quiz.

Simplify: $\frac{4+9+2}{3}$.

If you missed this problem, review Example 4.48.

BE PREPARED 5.14 Simplify: $4\,(8)+6\,(3)$.
If you missed this problem, review Example 2.8.

BE PREPARED 5.15 Convert $\frac{5}{2}$ to a decimal.
If you missed this problem, review Example 5.28.

One application of decimals that arises often is finding the *average* of a set of numbers. What do you think of when you hear the word *average*? Is it your grade point average, the average rent for an apartment in your city, the batting average of a player on your favorite baseball team? The average is a typical value in a set of numerical data. Calculating an average sometimes involves working with decimal numbers. In this section, we will look at three different ways to calculate an average.

Calculate the Mean of a Set of Numbers

The mean is often called the arithmetic average. It is computed by dividing the sum of the values by the number of values. Students want to know the mean of their test scores. Climatologists report that the mean temperature has, or has not, changed. City planners are interested in the mean household size.

Suppose Ethan's first three test scores were $85, 88, \text{and } 94$. To find the mean score, he would add them and divide by 3.

$$\frac{85+88+94}{3}$$
$$\frac{267}{3}$$
$$89$$

His mean test score is 89 points.

The Mean

The **mean** of a set of n numbers is the arithmetic average of the numbers.

$$\text{mean} = \frac{\text{sum of values in data set}}{n}$$

HOW TO

Calculate the mean of a set of numbers.

Step 1. Write the formula for the mean
$$\text{mean} = \frac{\text{sum of values in data set}}{n}$$
Step 2. Find the sum of all the values in the set. Write the sum in the numerator.
Step 3. Count the number, n, of values in the set. Write this number in the denominator.
Step 4. Simplify the fraction.
Step 5. Check to see that the mean is reasonable. It should be greater than the least number and less than the greatest number in the set.

EXAMPLE 5.49

Find the mean of the numbers $8, 12, 15, 9, \text{and } 6$.

Solution

Write the formula for the mean:	$\text{mean} = \frac{\text{sum of all the numbers}}{n}$
Write the sum of the numbers in the numerator.	$\text{mean} = \frac{8+12+15+9+6}{n}$
Count how many numbers are in the set. There are 5 numbers in the set, so $n = 5$.	$\text{mean} = \frac{8+12+15+9+6}{5}$
Add the numbers in the numerator.	$\text{mean} = \frac{50}{5}$
Then divide.	$\text{mean} = 10$
Check to see that the mean is 'typical': 10 is neither less than 6 nor greater than 15.	The mean is 10.

TRY IT 5.97 Find the mean of the numbers: $8, 9, 7, 12, 10, 5$.

TRY IT 5.98 Find the mean of the numbers: $9, 13, 11, 7, 5$.

EXAMPLE 5.50

The ages of the members of a family who got together for a birthday celebration were $16, 26, 53, 56, 65, 70, 93,$ and 97 years. Find the mean age.

Solution

Write the formula for the mean:	$\text{mean} = \frac{\text{sum of all the numbers}}{n}$
Write the sum of the numbers in the numerator.	$\text{mean} = \frac{16+26+53+56+65+70+93+97}{n}$
Count how many numbers are in the set. Call this n and write it in the denominator.	$\text{mean} = \frac{16+26+53+56+65+70+93+97}{8}$
Simplify the fraction.	$\text{mean} = \frac{476}{8}$
	$\text{mean} = 59.5$

Is 59.5 'typical'? Yes, it is neither less than 16 nor greater than 97. The mean age is 59.5 years.

TRY IT 5.99 The ages of the four students in Ben's carpool are $25, 18, 21,$ and 22. Find the mean age of the students.

TRY IT 5.100 Yen counted the number of emails she received last week. The numbers were $4, 9, 15, 12, 10, 12,$ and 8. Find the mean number of emails.

Did you notice that in the last example, while all the numbers were whole numbers, the mean was 59.5, a number with one decimal place? It is customary to report the mean to one more decimal place than the original numbers. In the next example, all the numbers represent money, and it will make sense to report the mean in dollars and cents.

EXAMPLE 5.51

For the past four months, Daisy's cell phone bills were $42.75, $50.12, $41.54, $48.15. Find the mean cost of Daisy's cell phone bills.

Solution

Write the formula for the mean.	$\text{mean} = \frac{\text{sum of all the numbers}}{n}$
Count how many numbers are in the set. Call this n and write it in the denominator.	$\text{mean} = \frac{\text{sum of all the numbers}}{4}$
Write the sum of all the numbers in the numerator.	$\text{mean} = \frac{42.75+50.12+41.54+48.15}{4}$
Simplify the fraction.	$\text{mean} = \frac{182.56}{4}$
	$\text{mean} = 45.64$

Does $45.64 seem 'typical' of this set of numbers? Yes, it is neither less than $41.54 nor greater than $50.12.

The mean cost of her cell phone bill was $45.64

TRY IT 5.101 Last week Ray recorded how much he spent for lunch each workday. He spent $6.50, $7.25, $4.90, $5.30, and $12.00. Find the mean of how much he spent each day.

TRY IT 5.102 Lisa has kept the receipts from the past four trips to the gas station. The receipts show the following amounts: $34.87, $42.31, $38.04, and $43.26. Find the mean.

Find the Median of a Set of Numbers

When Ann, Bianca, Dora, Eve, and Francine sing together on stage, they line up in order of their heights. Their heights, in inches, are shown in Table 5.6.

Ann	Bianca	Dora	Eve	Francine
59	60	65	68	70

Table 5.6

Dora is in the middle of the group. Her height, 65″, is the *median* of the girls' heights. Half of the heights are less than or equal to Dora's height, and half are greater than or equal. The median is the middle value.



Median

The **median** of a set of data values is the middle value.

- Half the data values are less than or equal to the median.
- Half the data values are greater than or equal to the median.

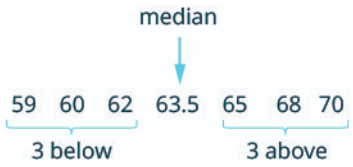
What if Carmen, the pianist, joins the singing group on stage? Carmen is 62 inches tall, so she fits in the height order between Bianca and Dora. Now the data set looks like this:

$$59, 60, 62, 65, 68, 70$$

There is no single middle value. The heights of the six girls can be divided into two equal parts.

59 60 62 | 65 68 70

Statisticians have agreed that in cases like this the median is the mean of the two values closest to the middle. So the median is the mean of 62 and 65, $\frac{62+65}{2}$. The median height is 63.5 inches.



Notice that when the number of girls was 5, the median was the third height, but when the number of girls was 6, the median was the mean of the third and fourth heights. In general, when the number of values is odd, the median will be the one value in the middle, but when the number is even, the median is the mean of the two middle values.

HOW TO

Find the median of a set of numbers.

Step 1. List the numbers from smallest to largest.
Step 2. Count how many numbers are in the set. Call this n.
Step 3. Is n odd or even?
- If n is an odd number, the median is the middle value.
- If n is an even number, the median is the mean of the two middle values.

EXAMPLE 5.52

Find the median of $12, 13, 19, 9, 11, 15,$ and 18.

Solution

List the numbers in order from smallest to largest.	9, 11, 12, 13, 15, 18, 19
Count how many numbers are in the set. Call this n.	$n = 7$
Is n odd or even?	odd
The median is the middle value.	median 9, 11, 12, **13,** 15, 18, 19 3 below 3 above
The middle is the number in the 4th position.	So the median of the data is 13.

TRY IT 5.103 Find the median of the data set: $43, 38, 51, 40, 46$.

TRY IT 5.104 Find the median of the data set: $15, 35, 20, 45, 50, 25, 30$.

EXAMPLE 5.53

Kristen received the following scores on her weekly math quizzes:

83, 79, 85, 86, 92, 100, 76, 90, 88, and 64. Find her median score.

Solution

Find the median of 83, 79, 85, 86, 92, 100, 76, 90, 88, and 64.	
List the numbers in order from smallest to largest.	64, 76, 79, 83, 85, 86, 88, 90, 92, 100
Count the number of data values in the set. Call this n.	$n = 10$
Is n odd or even?	even
The median is the mean of the two middle values, the 5th and 6th numbers.	64, 76, 79, 83, **85,** (5 numbers) **86,** 88, 90, 92, 100 (5 numbers)
Find the mean of 85 and 86.	$\text{mean} = \frac{85+86}{2}$
	$\text{mean} = 85.5$
	Kristen's median score is 85.5.

TRY IT 5.105 Find the median of the data set: 8, 7, 5, 10, 9, 12.

TRY IT 5.106 Find the median of the data set: 21, 25, 19, 17, 22, 18, 20, 24.

Identify the Mode of a Set of Numbers

The *average* is one number in a set of numbers that is somehow typical of the whole set of numbers. The mean and median are both often called the average. Yes, it can be confusing when the word average refers to two different numbers, the mean and the median! In fact, there is a third number that is also an average. This average is the **mode**. The mode of a set of numbers is the number that occurs the most. The **frequency**, is the number of times a number occurs. So the mode of a set of numbers is the number with the highest frequency.

Mode

The **mode** of a set of numbers is the number with the highest frequency.

Suppose Jolene kept track of the number of miles she ran since the start of the month, as shown in Figure 5.7.

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
				1 2 mi New Year's day	2	3 15 mi
4 8 mi	5	6 3 mi	7 8 mi	8	9 5 mi	10 8 mi
11	12	13	14	15	16	17

Figure 5.7

If we list the numbers in order it is easier to identify the one with the highest frequency.

$$2, 3, 5, 8, 8, 8, 15$$

Jolene ran 8 miles three times, and every other distance is listed only once. So the mode of the data is 8 miles.

HOW TO

Identify the mode of a set of numbers.

Step 1. List the data values in numerical order.
Step 2. Count the number of times each value appears.
Step 3. The mode is the value with the highest frequency.

EXAMPLE 5.54

The ages of students in a college math class are listed below. Identify the mode.

$18, 18, 18, 18, 19, 19, 19, 20, 20, 20, 20, 20, 20, 20, 21, 21, 22, 22, 22, 22, 22, 23, 24, 24, 25, 29, 30, 40, 44$

Solution

The ages are already listed in order. We will make a table of frequencies to help identify the age with the highest frequency.

Age	18	19	20	21	22	23	24	25	29	30	40	44
Frequency	4	3	7	2	5	1	2	1	1	1	1	1

Now look for the highest frequency. The highest frequency is 7, which corresponds to the age 20. So the mode of the ages in this class is 20 years.

TRY IT 5.107 The number of sick days employees used last year: $3, 6, 2, 3, 7, 5, 6, 2, 4, 2$. Identify the mode.

TRY IT 5.108 The number of handbags owned by women in a book club: $5, 6, 3, 1, 5, 8, 1, 5, 8, 5$. Identify the mode.

EXAMPLE 5.55

The data lists the heights (in inches) of students in a statistics class. Identify the mode.

56	61	63	64	65	66	67	67
60	62	63	64	65	66	67	70
60	63	63	64	66	66	67	74
61	63	64	65	66	67	67	

Solution

List each number with its frequency.

Number	56	60	61	62	63	64	65	66	67	70	74
Frequency	1	2	2	1	5	4	3	5	6	1	1

Now look for the highest frequency. The highest frequency is 6, which corresponds to the height 67 inches. So the mode of this set of heights is 67 inches.

TRY IT 5.109 The ages of the students in a statistics class are listed here: 19, 20, 23, 23, 38, 21, 19, 21, 19, 21, 20, 43, 20, 23, 17, 21, 21, 20, 29, 18, 28. What is the mode?

TRY IT 5.110 Students listed the number of members in their household as follows: 6, 2, 5, 6, 3, 7, 5, 6, 5, 3, 4, 4, 5, 7, 6, 4, 5, 2, 1, 5. What is the mode?

Some data sets do not have a mode because no value appears more than any other. And some data sets have more than one mode. In a given set, if two or more data values have the same highest frequency, we say they are all modes.

Use the Basic Definition of Probability

The probability of an event tells us how likely that event is to occur. We usually write probabilities as fractions or decimals.

For example, picture a fruit bowl that contains five pieces of fruit - three bananas and two apples.

If you want to choose one piece of fruit to eat for a snack and don't care what it is, there is a $\frac{3}{5}$ probability you will choose a banana, because there are three bananas out of the total of five pieces of fruit. The probability of an event is the number of favorable outcomes divided by the total number of outcomes.

$$\text{Probability of an event} = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}}$$

$$\text{Probability of choosing a banana} = \frac{3}{5} \begin{array}{l} \leftarrow \text{There are 3 bananas.} \\ \leftarrow \text{There are 5 pieces of fruit.} \end{array}$$

Probability

The **probability** of an event is the number of favorable outcomes divided by the total number of outcomes possible.

$$\text{Probability} = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}}$$

Converting the fraction $\frac{3}{5}$ to a decimal, we would say there is a 0.6 probability of choosing a banana.

$$\text{Probability of choosing a banana} = \frac{3}{5}$$
$$\text{Probability of choosing a banana} = 0.6$$

This basic definition of probability assumes that all the outcomes are equally likely to occur. If you study probabilities in a later math class, you'll learn about several other ways to calculate probabilities.

EXAMPLE 5.56

The ski club is holding a raffle to raise money. They sold 100 tickets. All of the tickets are placed in a jar. One ticket will be pulled out of the jar at random, and the winner will receive a prize. Cherie bought one raffle ticket.

ⓐ Find the probability she will win the prize.

ⓑ Convert the fraction to a decimal.

Solution

ⓐ

What are you asked to find?	The probability Cherie wins the prize.
What is the number of favorable outcomes?	1, because Cherie has 1 ticket.

Use the definition of probability.	$\text{Probability of an event} = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}}$
Substitute into the numerator and denominator.	$\text{Probability Cherie wins} = \frac{1}{100}$

ⓑ

Convert the fraction to a decimal.	
Write the probability as a fraction.	$\text{Probability} = \frac{1}{100}$
Convert the fraction to a decimal.	$\text{Probability} = 0.01$

TRY IT 5.111 Ignaly is attending a fashion show where the guests are seated at tables of ten. One guest from each table will be selected at random to receive a door prize. ⓐ Find the probability Ignaly will win the door prize for her table. ⓑ Convert the fraction to a decimal.

TRY IT 5.112 Hoang is among 20 people available to sit on a jury. One person will be chosen at random from the 20. ⓐ Find the probability Hoang will be chosen. ⓑ Convert the fraction to a decimal.

EXAMPLE 5.57

Three women and five men interviewed for a job. One of the candidates will be offered the job.

ⓐ Find the probability the job is offered to a woman.

ⓑ Convert the fraction to a decimal.

Solution

ⓐ

What are you asked to find?	The probability the job is offered to a woman.
What is the number of favorable outcomes?	3, because there are three women.
What are the total number of outcomes?	8, because 8 people interviewed.
Use the definition of probability.	$\text{Probability of an event} = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}}$
Substitute into the numerator and denominator.	$\text{Probability} = \frac{3}{8}$

ⓑ

Convert the fraction to a decimal.	

Write the probability as a fraction.	$\text{Probability} = \frac{3}{8}$
Convert the fraction to a decimal.	$\text{Probability} = 0.375$

TRY IT 5.113 A bowl of Halloween candy contains 5 chocolate candies and 3 lemon candies. Tanya will choose one piece of candy at random. (a) Find the probability Tanya will choose a chocolate candy. (b) Convert the fraction to a decimal.

TRY IT 5.114 Dan has 2 pairs of black socks and 6 pairs of blue socks. He will choose one pair at random to wear tomorrow. (a) Find the probability Dan will choose a pair of black socks (b) Convert the fraction to a decimal.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Mean, Median, and Mode (https://www.openstax.org/l/24meanmedmode)
Find the Mean of a Data Set (https://www.openstax.org/l/24meandataset)
Find the Median of a Data Set (https://www.openstax.org/l/24meddataset)
Find the Mode of a Data Set (https://www.openstax.org/l/24modedataset)

SECTION 5.5 EXERCISES

Practice Makes Perfect

Calculate the Mean of a Set of Numbers

In the following exercises, find the mean.

357. 3, 8, 2, 2, 5

358. 6, 1, 9, 3, 4, 7

359. 65, 13, 48, 32, 19, 33

360. 34, 45, 29, 61, and 41

361. 202, 241, 265, 274

362. 525, 532, 558, 574

363. 12.45, 12.99, 10.50, 11.25, 9.99, 12.72

364. 28.8, 32.9, 32.5, 27.9, 30.4, 32.5, 31.6, 32.7

365. Four girls leaving a mall were asked how much money they had just spent. The amounts were \$0, \$14.95, \$35.25, and \$25.16. Find the mean amount of money spent.

366. Juan bought 5 shirts to wear to his new job. The costs of the shirts were \$32.95, \$38.50, \$30.00, \$17.45, and \$24.25. Find the mean cost.

367. The number of minutes it took Jim to ride his bike to school for each of the past six days was 21, 18, 16, 19, 24, and 19. Find the mean number of minutes.

368. Norris bought six books for his classes this semester. The costs of the books were \$74.28, \$120.95, \$52.40, \$10.59, \$35.89, and \$59.24. Find the mean cost.

369. The top eight hitters in a softball league have batting averages of .373, .360, .321, .321, .320, .312, .311, and .311. Find the mean of the batting averages. Round your answer to the nearest thousandth.

370. The monthly snowfall at a ski resort over a six-month period was 60.3, 79.7, 50.9, 28.0, 47.4, and 46.1 inches. Find the mean snowfall.

Find the Median of a Set of Numbers

In the following exercises, find the median.

371. 24, 19, 18, 29, 21

372. 48, 51, 46, 42, 50

373. 65, 56, 35, 34, 44, 39, 55, 52, 45

374. 121, 115, 135, 109, 136, 147, 127, 119, 110

375. 4, 8, 1, 5, 14, 3, 1, 12

376. 3, 9, 2, 6, 20, 3, 3, 10

377. 99.2, 101.9, 98.6, 99.5, 100.8, 99.8

378. 28.8, 32.9, 32.5, 27.9, 30.4, 32.5, 31.6, 32.7

379. Last week Ray recorded how much he spent for lunch each workday. He spent $6.50, $7.25, $4.90, $5.30, and $12.00. Find the median.

380. Michaela is in charge of 6 two-year olds at a daycare center. Their ages, in months, are 25, 24, 28, 32, 29, and 31. Find the median age.

381. Brian is teaching a swim class for 6 three-year olds. Their ages, in months, are 38, 41, 45, 36, 40, and 42. Find the median age.

382. Sal recorded the amount he spent for gas each week for the past 8 weeks. The amounts were $38.65, $32.18, $40.23, $51.50, $43.68, $30.96, $41.37, and $44.72. Find the median amount.

Identify the Mode of a Set of Numbers

In the following exercises, identify the mode.

383. 2, 5, 1, 5, 2, 1, 2, 3, 2, 3, 1

384. 8, 5, 1, 3, 7, 1, 1, 7, 1, 8, 7

385. 18, 22, 17, 20, 19, 20, 22, 19, 29, 18, 23, 25, 22, 24, 23, 22, 18, 20, 22, 20

386. 42, 28, 32, 35, 24, 32, 48, 32, 32, 24, 35, 28, 30, 35, 45, 32, 28, 32, 42, 42, 30

387. The number of children per house on one block: 1, 4, 2, 3, 3, 2, 6, 2, 4, 2, 0, 3, 0.

388. The number of movies watched each month last year: 2, 0, 3, 0, 0, 8, 6, 5, 0, 1, 2, 3.

389. The number of units being taken by students in one class: 12, 5, 11, 10, 10, 11, 5, 11, 11, 11, 10, 12.

390. The number of hours of sleep per night for the past two weeks: 8, 5, 7, 8, 8, 6, 6, 6, 6, 9, 7, 8, 8, 8.

Use the Basic Definition of Probability

In the following exercises, express the probability as both a fraction and a decimal. (Round to three decimal places, if necessary.)

391. Josue is in a book club with 20 members. One member is chosen at random each month to select the next month's book. Find the probability that Josue will be chosen next month.

392. Jessica is one of eight kindergarten teachers at Mandela Elementary School. One of the kindergarten teachers will be selected at random to attend a summer workshop. Find the probability that Jessica will be selected.

393. There are 24 people who work in Dane's department. Next week, one person will be selected at random to bring in doughnuts. Find the probability that Dane will be selected. Round your answer to the nearest thousandth.

394. Monica has two strawberry yogurts and six banana yogurts in her refrigerator. She will choose one yogurt at random to take to work. Find the probability Monica will choose a strawberry yogurt.

395. Michel has four rock CDs and six country CDs in his car. He will pick one CD to play on his way to work. Find the probability Michel will pick a rock CD.

396. Noah is planning his summer camping trip. He can't decide among six campgrounds at the beach and twelve campgrounds in the mountains, so he will choose one campground at random. Find the probability that Noah will choose a campground at the beach.

397. Donovan is considering transferring to a 4-year college. He is considering 10 out-of state colleges and 4 colleges in his state. He will choose one college at random to visit during spring break. Find the probability that Donovan will choose an out-of-state college.

398. There are 258,890,850 number combinations possible in the Mega Millions lottery. One winning jackpot ticket will be chosen at random. Brent chooses his favorite number combination and buys one ticket. Find the probability Brent will win the jackpot. Round the decimal to the first digit that is not zero, then write the name of the decimal.

Everyday Math

399. Joaquin gets paid every Friday. His paychecks for the past 8 Fridays were $315, $236.25, $236.25, $236.25, $315, $315, $236.25, $393.75. Find the ⓐ mean, ⓑ median, and ⓒ mode.

400. The cash register receipts each day last week at a coffee shop were $1,845, $1,520, $1,438, $1,682, $1,850, $2,721, $2,539. Find the ⓐ mean, ⓑ median, and ⓒ mode.

Writing Exercises

401. Explain in your own words the difference between the mean, median, and mode of a set of numbers.

402. Make an example of probability that relates to your life. Write your answer as a fraction and explain what the numerator and denominator represent.